Two of theories should be contained here:

- Measure Theory;
- Real Analysis;

# Chapter 1 Measure

## 1.1 Construction and Extensions of Measures

### 1.1.1 Algebras, Semi-Algebras

**Algebras**

**Definition 1.1.1.1** Let $X$ be arbitrary set, then a collection $\mathcal{A}$ of subsets of $X$ is an algebra on $X$ if:

- $X, \emptyset \in \mathcal{A}$;(A I)
- $$A, B \in \mathcal{A} \Rightarrow \begin{cases} A \cap B \in \mathcal{A} & \text{(A II)} \\ A \cup B \in \mathcal{A} & \text{(A III)} \\ A - B \in \mathcal{A} & \text{(A IV)} \end{cases}$$

♠ **Proposition 1.1.1.2** For algebra axiom, define

$$B \in \mathcal{A} \Rightarrow X - B \in \mathcal{A} \qquad \text{(A iv)}$$

Then we have

$$\{(AI), (AII), (AIII), (AIV)\} \quad \text{equivalent to } \{(AI), (AII), (AII), (Aiv)\}$$

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∘ **Proof** ... □

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♠ **Proposition 1.1.1.3** TFAE:

- $X \in \mathcal{A}$;
- There exists a set $E \in \mathcal{A}$ which is called the **unit** of the algebra, if

$$A \cap E = A \qquad \forall A \in \mathcal{A}$$

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∘ **Proof** $\Rightarrow$ Choose $E = X$;
$\Leftarrow$ □

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And $\emptyset \in \mathcal{A}$ is not necessary. Notice

$$\emptyset = X - X \in \mathcal{A}$$

**Remark** Such algebra is actually a kind of boolean algebra. We have following correspondence

$$\begin{array}{cc} A \cup B & a \vee b \\ A \cap B & a \wedge b \\ X - A & \neg a \\ \emptyset & 0 \\ X & 1 \\ A \subseteq B & a \leq b \end{array}$$

### $\sigma$-Algebra

**Definition 1.1.1.4** An algebra of sets $\mathcal{A}$ is called a **$\sigma$-algebra** if for any sequence of sets $(A_n)_{n\in\omega}$, we have

$$A_n \in \mathcal{A} \Rightarrow \cup A_n \in \mathcal{A} \qquad \text{(SA I)}$$

Then

**Definition 1.1.1.5** A pair $(X, \mathcal{A})$ consisting of a set $X$ and a $\sigma$-algebra $\mathcal{A}$ is called a **measurable space**.

**Definition 1.1.1.6** Let $X$ be a set. For any family $\mathscr{F}$ of subsets of $X$, there exists a unique $\sigma$-algebra generated by $\mathscr{F}$, denoted by symbol

$$\sigma(\mathscr{F})$$

which is the smallest $\sigma$-algebra generated by $\mathscr{F}$.

♠ **Proposition 1.1.1.7** Let $X$ be a set.
Then, for any family of subsets of $X$, there exists a unique $\sigma$-algebra generated by $\mathscr{F}$.
In addition, there exists a unique algebra generated by $\mathscr{F}$.

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∘ **Proof** Such algebra is just the intersection of those $\sigma$ algebras, which contains $\mathscr{F}$. □

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### Morphisms

♠ **Proposition 1.1.1.8** Let $(A_\alpha)$ be a family of subsets of a set $X$, and letr $f : E \to X$ be an arbitrary mapping of a set $E$ to $X$. Then,

$$X - \cup_{\alpha\in\Lambda} A_\alpha = \cap \qquad X - \cap_{\alpha\in\Lambda} A_\alpha = \cup_{\alpha\in\Lambda}(X - A_\alpha)$$
$$f^{-1}(\cup_{\alpha\in\Lambda} A_\alpha) = \cup_{\alpha\in\Lambda} f^{-1}(A_\alpha) \qquad f^{-1}(\cap_{\alpha\in\Lambda} A_\alpha) = \cap_{\alpha\in\Lambda} f^{-1}(A_\alpha)$$

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∘ **Proof** ... □

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♠ **Proposition 1.1.1.9** Let $E, X$ be sets, with function

$$f : E \to X$$

- Let $\mathcal{A}$ be a $\sigma$-algebra of subsets of a set $X$,
  Then, the class $f^{-1}(\mathcal{A})$ of all sets of the form $f^{-1}(A)$, $A \in \mathcal{A}$, is a $\sigma$-algebra in $E$.
- Let $\mathcal{B}$ be a $\sigma$-algebra of subsets of a set $X$.
  Then, the class $\{A \subseteq X | f^{-1}(A) \in \mathcal{B}\}$ is a $\sigma$-algebra;
- Furthermore, for a class of subsets $\mathcal{F} \subseteq P(X)$

$$\sigma(f^{-1}\mathscr{F}) = f^{-1}(\sigma(\mathscr{F}))$$

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∘ **Proof**

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□

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This is actually the measure induced by function $f$. Could be seen as a pullback.

**Definition 1.1.1.10** Take the following notion of induced algebras:

$$f^{-1}\mathcal{A} \qquad f\mathcal{B}$$

**Borel $\sigma$-Algebra**

**Definition 1.1.1.11** The Borel $\sigma$-algebra of $\mathbb{R}^n$ is the $\sigma$-algebra

$$B(\mathbb{R}^n) = \sigma(Open(\mathbb{R}^n))$$

And take the following notion: $E \subseteq \mathbb{R}^n$. Then,

$$B(E) = \{E \cap B | B \in B(\mathbb{R}^n)\}$$

♠ **Proposition 1.1.1.12** The Borel $\sigma$-algebra of the real line is generated by any of the following classes of sets:

- The collection of all intervals;
- The collection of all intervals with rational endpoints;
- The collection of all rays of the form $(-\infty, c)$, where $c$ is rational;
- The collection of all rays of the form $(-\infty, c]$, where $c$ is rational;
- The collection of rays of the form $(c, \infty)$, where $c$ is rational;
- The collection of rays of the form $[c, \infty)$, where $c$ is rational;

Finally, the same is true if in place of rational numbers, one take points in anywhere dense set.

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∘ **Proof** Omitted. □

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♠ **Proposition 1.1.1.13** Consider the collection $\mathscr{F} = \{\{x\} | x \in X\}$, then

$$\sigma(\mathscr{F}) = \{A \subseteq X | A \text{ is at most countable}\} \cup \{A \subseteq X | X - A \text{ is at most countable}\}$$

Thus, $B(\mathbb{R}^n) < \sigma(\{\{x\} | x \in \mathbb{R}^n\})$.

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∘ **Proof** Easy to verify. □

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**Ring**

**Definition 1.1.1.14**

- Let $X$ be a set, with $S \subseteq P(X)$. Then, $S$ is a **semiring of sets** if

$$\emptyset \in S \qquad A \cap B \in S, \forall A, B \in S \qquad \sqcup_{i=1}^{n} C_i \in S, C_i \in S$$

- A family $R$ of subsets of a set $X$ is called a **ring** if

$$\emptyset \in R \qquad A, B \in R \Rightarrow A \cap B, A \cup B, A - B \in R$$

- A family $S$ of subsets of $X$ is called **semialgebra** if

$$\emptyset \in S \qquad A, B \in S \Rightarrow A \cap B \in S \qquad A, B \in S, A \subseteq B \Rightarrow B - A = \sqcup A_i,$$

- A ring is called a $\sigma$**-ring** if it is closed with respect to countable unions. i.e.

$$\cap A_n \in S$$

A ring is called a $\delta$**-ring** if it is closed with respect to countable intersections. i.e.

$$\cup A_n \in S$$

♠ **Proposition 1.1.1.15**

- Ring⇒ Semiring;
- Algebra⇒ Semialgebra;

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∘ **Proof**

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□

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♠ **Proposition 1.1.1.16** For any semising $S$, the collection

$$R = \{\cup_1^n X_i | X_i \in S\}$$

forms a ring $R$.

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∘ **Proof** □

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And

♠ **Proposition 1.1.1.17** For an algebra $\mathcal{A}$, TFAE:

- $\mathcal{A}$ is $\sigma$;
- $\mathcal{A}$ is $\delta$;

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∘ **Proof**

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□

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**Definition 1.1.1.18** Suppose $\mathscr{B}$ be a $\sigma$-algebra in space $X$, and any set $A \subseteq X$, the class

$$\mathscr{B}_A = \{B \cap A | B \in \mathscr{B}\}$$

The $\sigma$-algebra is called the **trace $\sigma$-algebra**.

### 1.1.2 Measure

**Additivity, Subadditivitty;Countable Additivity**

**Definition 1.1.2.1** Let $A$ be a set, take the notion of power set $P(A)$, and $A_i$ be a family of disjointed sets in $A$. Then, a **real-valued set function** $\mu : \mathcal{A} \to (-\infty, \infty)$ is defined on a class of sets $\mathcal{A}$ is called

- **additive** if

$$\mu(\cup_{i=1}^{n} A_i) = \sum \mu(A_i)$$

- **subadditive** if

$$\mu(\cup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mu(A_i)$$

- **countably additive** if

$$\mu(\cup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_i)$$

♠ **Proposition 1.1.2.2** Let $\mu$ be an additive real set function on an algebra/ ring of sets $\mathcal{A}$, then the following conditions are equivalent:

- The funciton $\mu$ is countably additive;
- The function $\mu$ is continuous at zero in following sense: For the following tower

$$A_n \in \mathcal{A} \qquad A_{n+1} \subseteq A_n$$

Then

$$\mu(\cap_{n=1}^{\infty} A_n) = 0$$

- The function $\mu$ is continuous from below. i.e. If $A_n \in \mathcal{A}$ are such that

$$A_n \subseteq A_{n+1}, \forall n \in \mathbb{N} \qquad \cup_{n=1}^{\infty} A_n \in \mathcal{A}$$

Then,

$$\mu(\cup_{n=1}^{\infty} A_n) = \lim_{n\to\infty} \mu(A_n)$$

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∘ **Proof**

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□

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Furthermore,

**Definition 1.1.2.3** Let $X$ be a set,
A function

$$\mu_0 : \mathcal{A} \to [0, +\infty]$$

- If $\mathcal{A} \subseteq P(X)$ be an ring of sets. $\mu_0$ is called **premeasure** if
  - $\mu_0(\emptyset) = 0$;
  - For a disjointed sequence $(A_n) \subseteq \mathcal{A}$,

$$\cup_{n=1}^{\infty} A_n \in \mathcal{A} \Rightarrow \mu_0(\sqcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu_0(A_n)$$

- If $\mathcal{A} \subseteq P(X)$ is aj algebra of sets.$\mu_0$ is called **measure** if it is countably additive.

♠ **Proposition 1.1.2.4** Let $\mu$ be a nonnegative additive set function on an algebra or ring $\mathcal{A}$:

- If $A, B \in \mathcal{A}, A \subseteq B$, then $\mu(A) \leq \mu(B)$;
- (Semiadditivity)For a collection

$$\mu(\cup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mu(A_i)$$

- (Countably-Semiadditivity) For a collection $\{A_n\}_{n\in\infty} \subseteq \mathcal{A}$, with $\cup_{n=1}^{\infty} A_n \in \mathcal{A}$, one has

$$\mu(\cup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

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∘ **Proof**

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□

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♠ **Proposition 1.1.2.5** Let $\mu$ be a nonnegative additive set function on an algebra or a ring $\mathcal{A}$:

- If $A, B \in \mathcal{A}, A \subseteq B, \mu(A) \leq \mu(B)$;
- For any collection $A_1, \ldots, A_n \in \mathcal{A}$, one has

$$\mu(\cup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mu(A_i)$$

- The function $\mu$ is countably additive precisely, when in addition to the additivity, it is countably subadditive in the following sense: For any sequence $\{A_n\} \subseteq \mathcal{A}$, with $\cup_{n=1}^{\infty} A_n \in \mathcal{A}$, one has

$$\mu(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

---

∘ **Proof**

- Easy to check;
- Obtained from (1)
- Similarly.

□

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♠ **Proposition 1.1.2.6** Let: $\mathcal{A}_0$ be a semialgebra.
Then, every additive set function $\mu$ on $\mathcal{A}_0$ uniquely extends to an additive set function on the algebra $\sigma(\mathcal{A}_0)$.
This extension is countably additive provided that $\mu$ is countably additive on $\mathcal{A}_0$.
The same is true for semiring and ring.

---

∘ **Proof**

- The extension: it is clear, that every element in $\mathcal{A}$ could be represented as the union of disjointed elements of $\mathcal{A}_0$. Now, set

$$\mu(A) := \sum_{i=1}^{n} \mu(A_i) \qquad A_i \in \mathcal{A}_0$$

Then, it is obviously additive. And such summation is idependent of choice of partion: If $B = \sqcup B_i$, then consider the equality

$$\mu(A_i) = \sum \mu(A_i \cap B_j) \qquad \mu(B_j) = \sum \mu(A_i \cap B_j)$$

- Now, the countable additivity: Suppose

$$A = \cup_{n=1}^{\infty} A_n \qquad A_n \cap A_k = \emptyset, n \neq k \qquad A_n \in \mathcal{A}$$

Then, for a family $A = \cup_{j=1}^{N} B_j$, $A_n = \cup_{i=1}^{N_n} B_{n,i}$, where $B_j, B_{n,j} \in \mathcal{A}_0$, set

$$C_{n,i,j} = B_{n,i} \cap B_j \in \mathcal{A}_0$$

Then, $\{c_{n,i,j}\}$ are pairwise disjoint, and then

$$B_j = \cup_{n=1}^{\infty} \cup_{i=1}^{N_n} \mu(C_{n,i,j}), B_{n,j} = \cup_{i=1}^{N_n} C_{n,i,j} \Rightarrow \mu(B_j) = \sum_{n=1}^{\infty} \sum_{i=1}^{N_n} \mu(C_{n,i,j}), \mu(B_{n,j}) = \sum_{i=1}^{N_n} \mu(C_{n,i,j})$$

Finally,

$$\mu(A) = \sum_{j=1}^{N} \sum_{n=1}^{\infty} \sum_{i=1}^{N_n} \mu(C_{n,i,j}) = \sum_{n=1}^{\infty} \sum_{j=1}^{N} \sum_{i=1}^{N_n} \mu(C_{n,i,j})$$

- Similarly for ring.

□

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**Probability**

Especially:

**Definition 1.1.2.7**

- A **countably additive measure** $\mu$ on a $\sigma$-algebra is called a **probability measure** if

$$\mu \geq 0 \qquad \mu(X) = 1$$

- Such $(X, \mathcal{A}, \mu)$ is called a **probability space**.

**Compact Classes, and Countable Additivity**

**Definition 1.1.2.8** A family $K$ of subsets of a set $X$ is called **compact** if, for any sequences $K_n$ of its elements,

$$\cap_{n=1}^{\infty} K_n \Rightarrow \cap_{n=1}^{N} K_n$$

♠ **Proposition 1.1.2.9** Let $\mu$ be a nonnegative additive set function on an algebra $\mathcal{A}$.
Suppose that: There exists a compact class $\mathcal{K}$ approximating $\mu$ in following sense: For every $A \in \mathcal{A}$, there exists

$$\exists K_\epsilon \in \mathcal{K}, A_\epsilon \subseteq K_\epsilon \subseteq A \qquad \mu(A - A_\epsilon) < \epsilon$$

Then, $\mu$ is countably additive. In particular, this is true if the compact class $\mathcal{K}$ is contained in $\mathcal{A}$ and for any $A \in \mathcal{A}$, one has the equality

$$\mu(A) = \sup_{K \subseteq A, K \in \mathcal{K}} \mu(K)$$

---

◦ **Proof** Suppose that the sets $A_n \in \mathcal{A}$ are decreasing, and

$$\cap_n A_n = \emptyset$$

Then, $\mu(A_n) \to 0$: Suppose that there exists a family of $K_n, B_n$ such that $B_n \subseteq K_n \subseteq A_n, \mu(A_n - B_n) < 2^{-n}\epsilon$.
By compactness, there exists some $N, \cap_{n=1}^{N} K_n = \emptyset \Rightarrow \cap_{n=1}^{N} B_n = \emptyset$. THen, one has

$$A_N = \cap_{n=1}^{N} A_n \subseteq \cup_{n=1}^{N} (A_n - B_n)$$
$$\Rightarrow \mu(A_n) \leq \sum_{n=1}^{N} \epsilon 2^{-n} \epsilon \leq \epsilon$$

□

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**Approxiamating Class**

**Definition 1.1.2.10** Let $m$ be a nonnegative function on a class $\mathscr{E} \subseteq P(X)$, and $\mathcal{P}$ be a class of subsets of $X$.
We say that $P$ is an approximating class for $m$, if for every $E \in \mathscr{E}$, every $\epsilon > 0$, there exists

$$P_\epsilon \in \mathcal{P}, E_\epsilon \in \mathcal{E}; E_\epsilon \subseteq P_\epsilon \subseteq E, |m(E) - m(E_\epsilon)| < \epsilon$$

**Approxiamating Class on Borel Set**

◇ **Lemma 1.1.2.11** Consider the class

$$\mathcal{A} = \{A \in \mathcal{B}(\mathbb{R}^n) | \forall \epsilon > 0, F \text{ closed}, U \text{ open}, F \subseteq A \subseteq U, \mu(U - F) < \epsilon\}$$

Then, $\mathcal{A} = B(\mathbb{R}^n)$

---

◦ **Proof** This is a Borel $\sigma$-algebra:

- Firstly, the closed subsets belongs to $\mathcal{A}$: Easy to check, just take the union of disks $A^\delta = \{x \in \mathbb{R}^n | dist(x, A) < \delta\}$, notice that

$$\lim_{k\to\infty}(A^{\frac{1}{k}} - A) = 0$$

  Thus, we could choose the approxiamating to $A$.

- Secondly, $\mathcal{A}$ is stable under compeletion:

$$F \subseteq A \subseteq U \Rightarrow U^c \subseteq A^c \subseteq F^c$$

- Thirdly: $\mathcal{A}$ is stable under countable union:

$$\mu(U_j - F_j) < \frac{\epsilon}{2^{j+1}} \Rightarrow \mu(U - F) \leq \sum \mu(U_j - F_j) < \frac{\epsilon}{2}$$

  But $F = \cup F_j$ may be not closed. Take a smaller $Z_k = \cup_1^k F_j$. Then, suppose $k$ is large enough, so $\mu(F - Z_k) < \frac{\epsilon}{2}$,

$$\mu(U - Z_k) \leq \mu(U - F) + \mu(F - Z_k) < \epsilon$$

Proved. □

---

♠ **Proposition 1.1.2.12** (Regularity of Borel Measure)Let $\mu$ be a nonnegative contably additive measure on the Borel $\sigma$-algebra $B(\mathbb{R}^n)$ in the space $\mathbb{R}^n$.
Then, for any Borel set $B \subseteq \mathbb{R}^n$, and $\epsilon > 0$, there exist an open subset $U_\epsilon$ and a compact set $K_\epsilon$ such that

$$K_\epsilon \subseteq B \subseteq U_\epsilon \qquad \mu(U_\epsilon - K_\epsilon) < \epsilon$$

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∘ **Proof** We've known, that for Borel set, we have approxiamation

$$F \subseteq B \subseteq U \qquad \mu(U - F) < \frac{\epsilon}{2}$$

where $F$ is closed. Then select $K_R = F \cap \bar{B}_R$, which is bounded and closed, so compact.
Notice that, $\lim_{R\to\infty} K_R = F$. Then,

$$\mu(F - K_R) \to 0$$

Choose such $R$ with $\mu(F - K_R) < \frac{\epsilon}{2}$, we obtrain

$$K_R \subseteq B \subseteq U$$

□

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★ **Corollary 1.1.2.13** $B(\mathbb{R}^n)$ is measurable.

### 1.1.3 Infinite Measure,$\sigma$-Finite Measures

Now, we will extend the value of measure to $[0, +\infty]$:

**Definition 1.1.3.1** Let

- $\mathcal{A}$ be a $\sigma$-algebra in a space $X$, and
- let $\mu$ be a set function on $\mathcal{A}$, with values in $[0, +\infty]$, and satisfies
  - $\mu(\emptyset) = 0$;
  - $\mu$ is countably additive

$$\mu(\cup A_j) = \sum \mu(A_j)$$

Then, $\mu$ is called a measure with values in $[0, \infty]$;

Especially, $\mu$ is called a $\sigma$**-finite** measure, if $X = \cup_{n=1}^{\infty} X_n, X_n \in \mathcal{A}, \mu(X_n) < \infty$;

◇ **Lemma 1.1.3.2** Let $\mathcal{R}$ be a ring of subsets of space $X$.

Let $\mu$ be a contably additive function on $\mathcal{R}$ with value $[0, +\infty]$ such that there exist sets $X_n \in \mathcal{R}$ with $X = \cup X_n, \mu(X_n) < \infty$.

Denote: $\mu_n$ the Lesbegue extension of measure $\mu$, regarded on the sets

$$S_n = \cup_{j=1}^{n} X_j$$

equipped with the algebra of sets consisting of the intersection of elements in $\mathcal{R}$ with $S_n$. Let $\mathcal{L}_{\mu_n}$ denote the class of all $\mu_n$-measurable sets. Let

$$\mathcal{A} := \{ A \subseteq X | A \cap S_n \in \mathcal{L}_{\mu_n}, \forall n \in \mathbb{N}, \}$$

Then, $\mathcal{A}$ is a ring with respect to countable intersection, and $\bar{\mu}$ is a $\sigma$-additive measure, whose restriction to those $S_n$ coincide with $\mu$.

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◦ **Proof** Just consider

$$\mu_n(A \cap S_n) = \sum \mu_n(A_i \cap S_n)$$

□

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♠ **Proposition 1.1.3.3** Let $\mu$ be a countably additive additive measure on an algebra $\mathcal{A}$, with values in $[0, \infty]$, then $A_\mu$ is a $\sigma$-algebra, $A \subseteq A_\mu$, and the function $\mu^*$ is a countably additive measure on $\mathcal{A}_\mu$ with values in $[0, +\infty]$ and coincides with $\mu$ on $\mathcal{A}$.

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◦ **Proof** ... □

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# 1.2 Outer Measure

## 1.2.1 Outer Measure and the Lebesgue Extension Measures

**Definition 1.2.1.1** Let: $\mu$ be a nonnegative set function $\mu$ that is defined a certain class $\mathcal{A}$ of subsets in a space $X$, and contains $X$ itself, then the formula

$$\mu^*(A) = \inf\{ \sum_{n=1}^{\infty} \mu(A_n) | A_n \in \mathcal{A}, A \subseteq \cup_{n=1}^{\infty} A_n \}, A \subseteq X$$

Then, $\mu^* : P(X) \to [0, +\infty]$.

**Definition 1.2.1.2** Suppose that $\mu$ is nonnegative set function on domain $\mathcal{A} \subseteq 2^X$.

A set $A$ is called $\mu$**-measurable** if for any $\epsilon > 0$, there exists $A_\epsilon \in \mathcal{A}$, such that

$$\mu^*(A \Delta A_\epsilon) < \epsilon$$

Take the notion of $\mu$-measurable subsets $A_\mu$.

◇ **Lemma 1.2.1.3** Let $\mu$ be a nonnegative set function on a class $\mathcal{A}$, then the function $\mu^*$ is contably subadditive. i.e.

$$\mu^*(\cup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$$

for any sets $A_n$.

∘ **Proof** ... □

◇ **Lemma 1.2.1.4** In the situation of previous lemma, for any sets $A$ and $B$, such that $\mu^*(B) < \infty$, one has the equality

$$|\mu^*(A) - \mu^*(B)| \leq \mu^*(A \Delta B)$$

∘ **Proof** ... □

♠ **Proposition 1.2.1.5** Let $\mu$ be a nonnegative countably additive set function on an algebra $\mathcal{A}$, then:

- One has $\mathcal{A} \subseteq \mathcal{A}_\mu$, and the outer measure $\mu^*$ coincides with $\mu$ on $\mathcal{A}$;
- The collection $\mathcal{A}_\mu$ of all $\mu$-measurable sets is a $\sigma$-algebra, nad the striction of $\mu^*$ to $\mathcal{A}_\mu$ is countably additive;
- The function $\mu^*$ is a unique nonnegative countably additive extension of $\mu$ to the $\sigma$-algebra $\sigma(\mathcal{A})$ generated by $\mathcal{A}$, and a unique nonnegative countably additive extension of $\mu$ to $\mathcal{A}_\mu$.

∘ **Proof** ... □

★ **Corollary 1.2.1.6** Let $\mu$ be a nonnegative countably additive set function on an algebra $\mathcal{A}$.
A set $A$ is $\mu$-measurable precisely when there exist two sets $A', A'' \in \sigma(\mathcal{A})$ such that

$$A' \subseteq A \subseteq A'' \qquad \mu^*(A'' - A') = 0$$

Moreover, one can take for $A'$ a set of the form $\cup_{n=1}^{\infty} \cap_{k=1}^{\infty} A_{n,k}, A_{n,k} \in \mathcal{A}$, and for $A''$ a set of the form $\cap_{n=1}^{\infty} \cup_{k=1}^{\infty} B_{n,k}, B_{n,k} \in \mathcal{A}$.

∘ **Proof** ... □

★ **Corollary 1.2.1.7** ...

∘ **Proof** ... □

**Collection $\mathcal{A}_\mu$**

♠ **Proposition 1.2.1.8** Let $\mu$ be a nonnegative countably additive measure on an algebra $\mathcal{A}$. Then, TFAE:

- $A \in \mathcal{A}_\mu$;
- $\mu^*(A) + \mu^*(X - A) = \mu^*(X)$;
- $\mu^*(E \cap A) + \mu^*(E - A) = \mu^*(E), E \subseteq X$.

∘ **Proof**

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□

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**Countable Additivity**

♠ **Proposition 1.2.1.9** Let $\mu$ be a nonnegative measure on a $\sigma$-algebra $\mathcal{A}$, suppose that sets $\mathcal{A}_n$ are such that $A_n \subseteq A_{n+1}$ for all $n \in \mathbb{N}$, then, one has

$$\mu^*(\cup A_n) = \lim \mu^*(A_n)$$

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∘ **Proof** ... □

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**Compeleteness**

**Definition 1.2.1.10** ...

## 1.2.2 Carathedory Outer Measure

**Definition 1.2.2.1** A set function $m : P(X) \to [0, +\infty]$ defined on the class of all subsets of a set, and taking values in $[0, +\infty]$ is called **outer measure** on $X$, if:

- $m(\emptyset) = 0$;
- $m(A) \leq m[B], A \subseteq B$;
- $m(\cup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} m(A_n)$;

Especially,

**Definition 1.2.2.2** Let $m$ be a set function with values in $[0, +\infty]$ defined on the class of all subsets of a space $X$, such that $m(\emptyset) = 0$.

A set $A \subseteq X$ is called **Caratheodory measurable** with respect to $m$, if for every $E \subseteq X$, one has the equality

$$m(E \cap A) + m(E - A) = m(E)$$

The class of all Caratheodory $m$-measurable sets is denoted by $\mathfrak{M}_m$.

♠ **Proposition 1.2.2.3** Let $m$ be a set function with values in $[0, +\infty]$ on the class of all sets in a space $X$, such that $m(\emptyset) = 0$, then

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- 
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∘ **Proof** ... □

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## 1.3 Lesbegue Measure

### 1.3.1 Lesbegue Measure

**Definition 1.3.1.1**

- Consider: The **cube** $I$ in $\mathbb{R}^n$ of form $[a,b]^n$
- Then, the algebra

$$\mathcal{A}_0 = \{\cup_j = 1^m I_j | m \in \mathbb{N}, I_j \text{ is cube}\}$$

- Volumes:

$$\lambda_n^0(\prod[a_i, b_i)) := \prod(b_i - a_i)$$

Then, by finite additivity, we could extend it to the whole $\mathcal{A}_0$.
Especially: Suppose $\sqcup_{j=1}^{\infty} A_j \in \mathcal{A}_0$, then

...

Thus, this is a premeasure

- According the construction by Caratheodory, we got an outer measure $\lambda_n^*$. This is the **Lesbegue $\sigma$-Algebra** on $I$.
- Finally, define:

...

By the increasing cubes,

### 1.3.2 Lesbegue-Stieltjes Measures

**Definition 1.3.2.1** (Distribution Function)

- Suppose $\mu$ be a nonnegative Borel measure on $\mathbb{R}$. Then, the function

$$t \mapsto F(t) = \mu((-\infty, t))$$

is bounded, nondecreasing

- Such function is often called the **distribution function**.

▶ ***Theorem*** Let $F$ be a bounded, nondecreasing, left continuous function, with

$$\lim_{t \to -\infty} F(t) = 0$$

Then, there exists a unique nonnegative Borel measure on $\mathbb{R}$ such that

$$F(t) = \mu((-\infty, t)) \qquad \forall t \in \mathbb{R}$$

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∘ **Proof** ... □

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## 1.4 Monotone and $\sigma$-additive classes of sets

## 1.5 Souslin Sets, and $A$-Operations

**Definition 1.5.0.1** ...

# Chapter 2 The Lesbegue Integral

## 2.1 Measurable Functions

**Definition 2.1.0.1** Let $(X, \mathcal{A})$ be a measurable space.
Then, a function $f : X \to \mathbb{R}$ is called **measurable** if

$$\{x | f(x) < c\} \subseteq X \in \mathcal{A} \qquad \forall c \in \mathbb{R}$$

♠ **Proposition 2.1.0.2** TFAE:

- A function $f$ is **measurable** with respect to a $\sigma$-algebra $\mathcal{A}$
- $f^{-1}(B) \in \mathcal{A}, \forall B \in \mathscr{B}(\mathbb{R})$.

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∘ **Proof** ... □

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**Definition 2.1.0.3** Define: The **indicator**

$$1_A(x) := \begin{cases} 1 & x \in A_i \\ 0 & Elsewise \end{cases}$$

It is easy to verify, $1_A$ is measurable.

Now, take the following notion

$$f^+(x) := \max(f(x), 0) \qquad f^-(x) := \max(-f(x), 0)$$

Then,

♠ **Proposition 2.1.0.4** TFAE: For function $f : \mathcal{A} \to \mathbb{R}$

- $f$ is measurable;
- $f^+, f^-$ are measurale;

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∘ **Proof** ... □

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♠ **Proposition 2.1.0.5** TFAE For function $f : (X, \mathcal{A}) \to (\mathbb{R}, B(\mathbb{R}))$,

- $f$ is measurable;
- $f^{-1}(B) \in \mathcal{A}$, $B \in \mathcal{B}(\mathbb{R})$;

---

∘ **Proof** ... □

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### 2.1.1 Mesurable Function beetween Measurable Spaces

**Definition 2.1.1.1** Let $(X_1, \mathcal{A}_1), (X_2, \mathcal{A}_2)$ be two spaces with $\sigma$-algebras.
A mapping $f : X_1 \to X_2$ is called **measurable** with respect to the pari $(\mathcal{A}_1, \mathcal{A}_2)$ if

$$B \in \mathcal{A}_2 \Rightarrow f^{-1}(B) \in \mathcal{A}_1$$

♠ **Proposition 2.1.1.2** Suppose that the function $f, g, f_n : X \to \mathbb{R}$, where $n \in \mathbb{N}$, are measurable with respect to a $\sigma$-algebra $\mathcal{A}$, then

- The function $\varphi \circ f$ is measurable with respect to $\mathcal{A}$ for Borel function

  $$\varphi : \mathbb{R} \to \mathbb{R}$$

  in particular, this is true if $\varphi$ is continuous.
- The function

  $$\alpha f + \beta g$$

  is measurable with respect to $\mathcal{A}$, for every $\alpha, \beta \in \mathbb{R}$;
- The function $fg$ is measurable with respect to $\mathcal{A}$;
- If $g(x) \neq 0$, the function $\frac{f}{g}$ is measurable with respect to $\mathcal{A}$;
- If there exists a finite limit $f_0(x) = \lim_{n\to\infty} f_n(x)$ for all $x$, then the function $f_0$ is measurable with respect to $\mathcal{A}$;
- If the functions $\sup_n f_n(x)$ and $\inf_n f_n(x)$ are finite for all $x$, then they are measurable with respect to $\mathcal{A}$.

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∘ **Proof**

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□

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♠ **Proposition 2.1.1.3** ...

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∘ **Proof** ... □

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**$\mu$-Measurable**

**Definition 2.1.1.4** Let $(X, \mathcal{A}, \mu)$ be a measure space.
A real function $f$ on $X$ is called **$\mu$-measurable** if it is measurable with respect to the $\sigma$-algebra $\mathcal{A}_\mu$ of all $\mu$-measurable sets.
In addition, we agree to call $\mu$-measurable aloso any function $f$ that is defined and $\mathcal{A}_\mu$-measuralbe on $X - Z$, where $Z$ is a set of $\mu$-measure zero.
The set of all $\mu$-measurable function is denoted by $\mathcal{L}^0(\mu)$.

♠ **Proposition 2.1.1.5** Let $\mu$ be a nonnegative measure on a $\sigma$-algebra $\mathcal{A}$.

Then, for every $\mu$-measurable function $f$, one can find a set $Y \in \mathcal{A}$ and a function $g$ measurable with respect to $\mathcal{A}$ such that

$$f(x) = g(x) \qquad \forall x \in Y, \mu(X - Y) = 0$$

---

∘ **Proof** ... □

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## 2.1.2 Convergence, Almost Everywhere

**Definition 2.1.2.1**

- Some property for points in $X$ is fullfilled **$\mu$-almost everywhere** on $X$ if the set $Z$ of all points in $X$ that do not have this property
  - belongs to $A_\mu$, and
  - has measure zero with respect to $\mu$.

  Denoted by 'a.e.'
- $f = g$ a.e. states, that

$$\{x | f(x) = g(x)\} \in \mathcal{A}, \mu\{x | f(x) = g(x)\} = 0$$

▶ ***Theorem*** (Egoroff) Let $(X, \mathcal{A}, \mu)$ be a space with a finite nonnegative measure $\mu$, and let $\mu$-measurable function $f_n$ be such that $\mu$-almost everywhere, there is a finite limit

$$f(x) = \lim_{n\to\infty} f_n(x)$$

Then for every $\epsilon > 0$, there exists a set $X_\epsilon \in \mathcal{A}$, such that

- $\mu(X - X_\epsilon) < \epsilon$;
- the function $f_n \overset{uniformly}{\to} X_\epsilon$ on $X_\epsilon$.

---

∘ **Proof** Define:

$$X_n^m := \cap_{i\geq n}\{x | |f_i(x) - f(x)| < \frac{1}{m}\} \in \mathcal{A}_\mu$$

Then, $X_n^m \subseteq X_{n+1}^m$. And $\cup_{n=1}^\infty X_n^m = X$: This is because that, for every $y \in X$, and fixed $m \in \mathbb{Z}^+$, because

$$\lim_{n\to\infty} f_n(y) = f(y)$$

there exists large enough $n$ such that $y \in X_n^m$.
Thus, for each $\epsilon > 0$, there exists a number $k(m)$ such that

$$\mu(X - X_{k(m)}^m) < \epsilon 2^{-m}$$

Now, let $X_\epsilon = \cap_{m=1}^\infty X_{k(m)}^m$, we obtain

$$\mu(X - X_\epsilon) \leq \sum \mu(X - X_{k(m)}^m) \leq \epsilon \sum_m 2^{-m} = \epsilon$$

□

---

**Definition 2.1.2.2** Suppose we are given a measure space $(X, \mathcal{A}, \mu)$ with a finite measure $\mu$, and a sequence of $\mu$-measurable functions:

- The sequence $\{f_n\}$ is called **fundamental** or **Cauchy** if, for every $c > 0$, one has

$$\lim_{N\to\infty} \sup_{n,k\geq N} \mu(x||f_n(x) - f_k(x)| \geq c) = 0$$

- The sequence $\{f_n\}$ is said to **converge** in measure to a $\mu$-measurable function $f$ if, for every $c > 0$, one has

$$\lim_{n\to\infty} \mu(x||f(x) - f_n(x)| \geq c) = 0$$

Notice, $\lim_{N\to\infty} \sup_{n,k\geq N} \mu(x||f_n(x) - f_k(x)| \geq c)$ is well-defined, because it is a descending sequence.

♠ **Proposition 2.1.2.3** Let $(X, \mathcal{A}, \mu)$ be a space with a finite measure($\mu(X) < \infty$)

- If a sequence of $\mu$-measurable functions $f_n$ converges to $f$ in measure $\mu$, then there exists its subsequence $f_{n_k} \to f$ almost everywhere.
- If a sequence of $\mu$-measurable functions $f_n$ is fundamental in measure $\mu$, then it cconverges in measure $\mu$ to some measurable function $f$.

---

∘ **Proof**

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□

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★ **Corollary 2.1.2.4** Let $\mu$ be a finite measure and let two sequences of $\mu$-measurable function $f_n, g_n$ converge in measure $\mu$ to functions $f$ and $g$.

Suppose that $\Psi$ is a continuous function on $Y \subseteq \mathbb{R}^2$, such that

$$(f(x), g(x)) \in Y \qquad (f_n(x), g_n(x)) \in Y$$

Then, $\Psi(f_n, g_n) \to \Psi(f, g)$.

---

∘ **Proof** ... □

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★ **Corollary 2.1.2.5** For such $f_n, g_n, f, g$,

$$f_n g_n \to fg \qquad \alpha f_n + \beta g_n \to \alpha f + \beta g$$

---

∘ **Proof** ... □

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**Approxiamation Extension from Compact**

◇ **Lemma 2.1.2.6** ...

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∘ **Proof** ... □

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♠ **Proposition 2.1.2.7** ...

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∘ **Proof** ... □

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▶ ***Theorem*** (Lusin's Theorem) A function $f$ on an interval $I$, with Lesbegue measure is measurable precisely when for each $\epsilon > 0$, there exists a continuous function

$$f_\epsilon : \mathbb{R} \to \mathbb{R}$$

and a compact set $K_\epsilon$ such that

$$\mu(I - K_\epsilon) < \epsilon \qquad f = f_\epsilon \text{ on } K_\epsilon$$

---

∘ **Proof** ... □

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## 2.2 Integrations

### 2.2.1 Simple Functions

**Definition 2.2.1.1**

- Let $(X, \mathcal{A})$ be a measurable space.
  A function

$$f : X \to \mathbb{R}$$

  is called **simple function**, if $f$ is measurable, and $f(X)$ is a finite subset.

♠ **Proposition 2.2.1.2** TFAE:

- $f$ is a simple function;
- $f = \sum a_i 1_{A_i}, \cup A_i \subseteq A$;
- $f = \sum a_i 1_{A_i}. \sqcup A_i \subseteq A$;

---

∘ **Proof** ... □

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**Definition 2.2.1.3** ...

◇ **Lemma 2.2.1.4** Let $\{f_n\}, \{g_n\}$ be tow sequences of simple functions, that are:

- mean fundamental($L^1$ Cauchy)
- Almost everywhere to one and the same function $f$.

Then, $f_n$ and $g_n$ converge to the same value.

---

∘ **Proof** ... □

◇ **Lemma 2.2.1.5** ...

∘ **Proof** ... □

**Definition 2.2.1.6** ...

## 2.2.2 Integral of Simple Functions

**Definition 2.2.2.1** Let $(X, \mathcal{A}, \mu)$ be a space with finite nonnegative measure.

For any simple function $f$ on $X$, that assmues finitely many values $c_i$ on disjoint sets $A_i$, the **Lesbegue integral** of $f$ with respect to $\mu$ is defined by

$$\int_X f(x)\mu(dx) := \sum_i c_i \mu(A_i)$$

**Fundamental/Mean Fundamental/Fundamental in $L^1(\mu)$**

**Definition 2.2.2.2** ...

◇ **Lemma 2.2.2.3** The Lesbegue integral on simple functions enjoys the following properties:

- 
- 
- 

∘ **Proof** ... □

★ **Corollary 2.2.2.4** ...

∘ **Proof** ... □

♠ **Proposition 2.2.2.5** ...

∘ **Proof** ... □

## 2.2.3 Lesbegue Integration

**Definition 2.2.3.1** ...

**Basic Properties**

**Chebyshev's Theorem**

▶ ***Theorem*** (Chebyshev/Markov Inequality)

$$\mu(\{x||f(x)| \geq R\}) \leq \frac{1}{R}\int_X |f(x)|\mu(dx) \qquad R > 0$$

---

∘ **Proof** ... □

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★ **Corollary 2.2.3.2**

$$\mu(\{|f| \geq \lambda\}) \leq \frac{\|f\|_{L^p}^p}{\lambda^p}$$

---

∘ **Proof** ... □

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**Covergence Theorem:Lesbegue/Levi/Fatou**

▶ ***Theorem*** (Lesbegue Dominated Covergence Theorem) Suppose that $\mu$-integrable functions $f_n$ converge almost everywhere to a function $f$.

If there exists a $\mu$-integrable function $\Phi$, such that

$$|f_n(x)| \leq \Phi(x), \forall n$$

Then,

- the function $f$ is integrable, and

$$\int_X f(x)\mu(dx) = \lim_{n\to\infty} f_n(x)\mu(dx)$$

- In addition

$$\lim_{n\to\infty}\int_X |f(x) - f_n(x)|\mu(dx) = 0$$

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∘ **Proof** ... □

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### 2.2.4 Criteria of Integrability

▶ ***Theorem*** ...

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∘ **Proof** ... □

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## 2.3 Space $L^1$

### 2.3.1 Completeness

**Definition 2.3.1.1**

- A sequence of function $f_n$ that are integrable with respect to a measure $\mu$, is calledd **fundamental**
-

## 2.4 Riemann Integral

# Chapter 3 Operations on Measures and Functions

## 3.1 Decompositikon on Measures, and Functions

# Chapter 4 Integral and Derivative

## 4.1 Differentiability of Functions on the Real Line

# Chapter 5 Borel,Baire and Souslin Sets

# Chapter 6 Measures on Topological Spaces

## 6.1 Meaasures on Manifolds

# Chapter 7 Weak Convergence of Measure

# Chapter 8 Transformations of Measures, and Isomorphisms

# Chapter 9 Conditional Measures and Conditional Expectation